

Prior Sensitivity in Bayesian Logistic Regression: A Small-Sample Medical Study

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Abstract

In Bayesian logistic regression, the choice of prior is inconsequential when data are plentiful but decisive when they are scarce. We demonstrate this using the `birthwt` dataset ($n = 189$), comparing three priors (diffuse $\mathcal{N}(0, 100)$, weakly informative $\mathcal{N}(0, 2.5)$, and a clinically informative prior) across the full sample and subsamples of $n \in \{20, 40, 80, 189\}$. At full data all three are interchangeable (AUC $\in [0.746, 0.748]$; LOOIC difference < 4). Under small samples they are not: at $n = 20$ the diffuse prior collapses under complete separation ($\hat{\beta}_{\text{smoke1}} = 203$) while the regularizing priors stay stable. Crucially, how quickly a coefficient escapes prior sensitivity is governed not by sample size but by the number of positive cases for that predictor: smoking stabilizes by $n = 40$, whereas hypertension (only 6.3% of patients) remains unstable until $n \approx 80$. For small clinical datasets the recommendation is concrete: prefer a weakly informative prior to a diffuse one, and treat rare predictors with particular caution, since they are the last to shed the prior’s influence.

1 Introduction

In Bayesian analysis the prior’s influence grows as data shrink: when samples are scarce it can substantially shape, and even distort, the posterior, a concern that is especially acute in clinical research. This report presents a prior sensitivity analysis of Bayesian logistic regression using the `birthwt` dataset ($n = 189$), comparing three priors (diffuse $\mathcal{N}(0, 100)$, weakly informative $\mathcal{N}(0, 2.5)$, and a clinically informative prior) refitted across $n \in \{20, 40, 80, 189\}$. We find that sensitivity is governed not by sample size alone but by predictor rarity, with direct implications for clinical datasets where the most important risk factors are often uncommon. Code is available at [GitHub Repository](#).

2 Data

2.1 Dataset Overview

The `birthwt` dataset ($n = 189$) was collected at Baystate Medical Center, Springfield, MA (1986). The binary outcome is low birth weight (< 2.5 kg), with 130 normal and 59 low birth weight cases (31.2% positive rate). All analyses use `stan.glm` (`rstanarm`), fit with 4 chains $\times$ 1000 iterations (500 warmup, 500 sampling); $\hat{R}$ and effective sample size were monitored at all n .

2.2 Predictors

Predictors include maternal age, pre-pregnancy weight, race, smoking status, prior preterm labor, hypertension history (`ht`), uterine irritability (`ui`), and first-trimester physician visits. Among the binary predictors, positive cases are sparse and uneven: 74/189 for smoking, 28 for uterine irritability, and only 12 for hypertension, a spread that becomes consequential under small samples.

3 Methods

3.1 Prior Specification

Three prior configurations were evaluated:

Model	Coefficients	Intercept
Diffuse	$\mathcal{N}(0, 100)$	$\mathcal{N}(0, 100)$
Weak	$\mathcal{N}(0, 2.5)$	$\mathcal{N}(0, 10)$
Informative	smoke, ht: $\mathcal{N}(1, 0.5)$; others $\mathcal{N}(0, 1)$	$\mathcal{N}(0, 5)$

The informative prior encodes clinical knowledge: $\mathcal{N}(1, 0.5)$ for **smoke** and **ht** corresponds to $\text{OR} \approx 2.7$ (± 2 SD: $\text{OR} \in [1.4, 7.4]$), in the expected direction given published risk estimates [1, 2]. All other predictors use $\mathcal{N}(0, 1)$.

3.2 Analyses

We conduct three analyses: (1) full-data prior comparison via posterior coefficients, LOO-CV, and AUC; (2) a subsampling experiment refitting all models at $n \in \{20, 40, 80, 189\}$ to track convergence across sample sizes, including a 100-draw repeated subsampling robustness check; and (3) predictive performance assessed via LOOIC and Pareto $\hat{k}$ diagnostics. Convergence was verified via $\hat{R} \leq 1.002$ and trace plots.

4 Results

4.1 Full Data ($n = 189$): Priors Have Little Effect

At $n = 189$, all three models produce nearly identical posterior estimates and converge well. Table 1 reports posterior means for the two clinically relevant predictors, and Figure 1 shows the full coefficient comparison across all predictors.

Variable	Diffuse		Weak		Informative	
	Mean	95% CI	Mean	95% CI	Mean	95% CI
smoke1	0.983	[0.21, 1.83]	0.933	[0.13, 1.74]	0.915	[0.31, 1.52]
ht1	1.994	[0.63, 3.50]	1.827	[0.49, 3.30]	1.311	[0.51, 2.13]

Table 1: Posterior means and 95% credible intervals (log-odds) for the two clinically targeted predictors at $n = 189$.

The informative prior noticeably shrinks **ht1** (from 1.994 to 1.311, $\text{OR} \approx 7.3$ to 3.7), reflecting prior pull even at full data. Table 2 reports LOO and AUC for all three models; the differences are negligible, confirming that prior choice has little practical effect at $n = 189$.

Model	LOOIC	AUC
Diffuse	223.85	0.746
Weak	223.32	0.747
Informative	220.43	0.748

Table 2: LOO model comparison and AUC at full data ($n = 189$). Lower LOOIC indicates better fit.

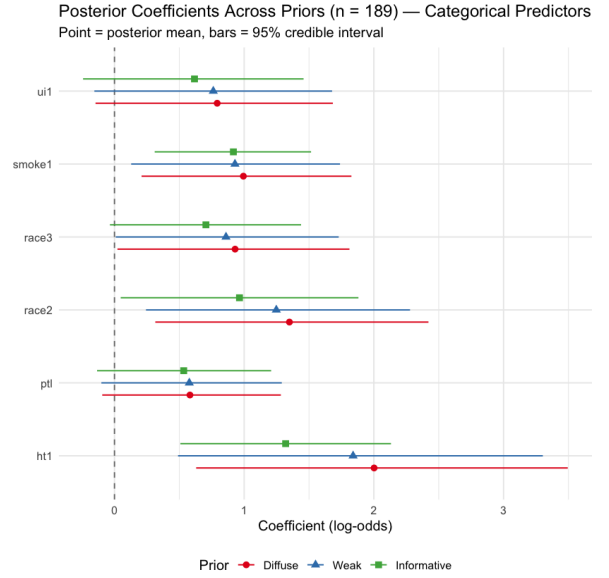
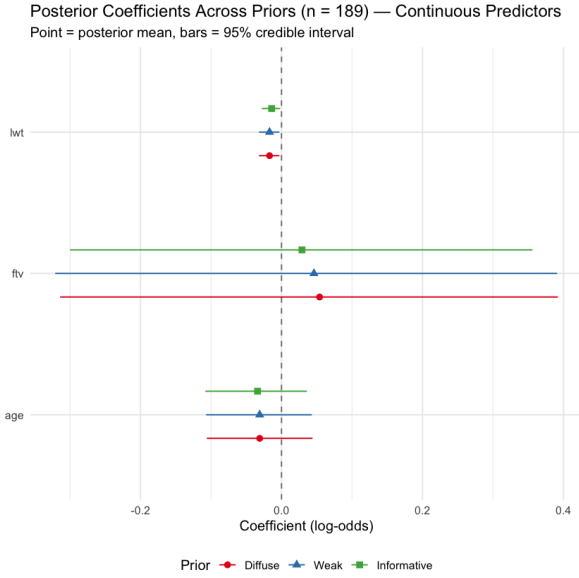


Figure 1: Posterior coefficients across priors at $n = 189$. Left: continuous predictors. Right: categorical/binary predictors. Points = posterior mean; bars = 95% credible interval.

4.2 Subsampling: Prior Sensitivity at Small n

To examine how prior influence changes with sample size, we refit all three models at $n \in \{20, 40, 80, 189\}$. We focus on **smoke1** and **ht1**, the two predictors for which the informative prior encodes directional knowledge, as these show the largest divergence across priors at small n (Figure 2).

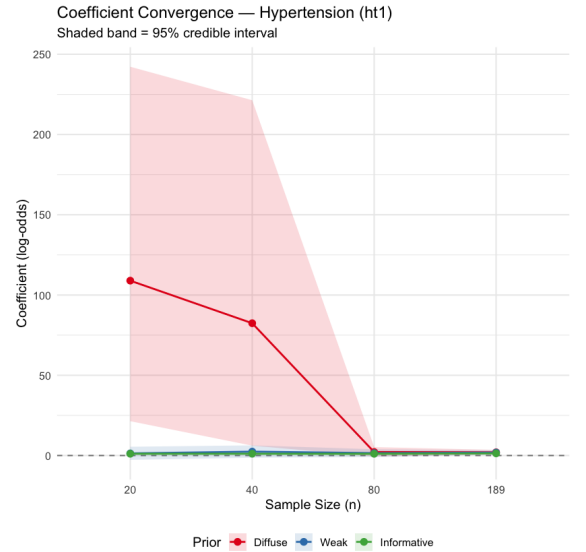
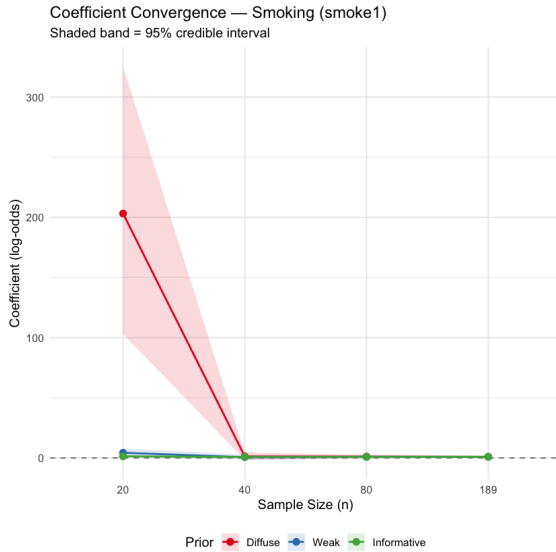


Figure 2: Posterior mean convergence. Left: **smoke1**. Right: **ht1**. Shaded bands = 95% credible intervals.

Under the diffuse prior, complete separation produces implausible estimates at $n = 20$ ($\hat{\beta}_{\text{smoke1}} = 203$, $\hat{\beta}_{\text{ht1}} = 109$). **smoke1** recovers by $n = 40$, but **ht1** remains unstable ($\hat{\beta} = 82.4$): hypertension affects only 12 of 189 patients (6.3%), so small subsamples rarely contain enough **ht** = 1 cases to resolve separation. By $n = 80$, all models stabilize. The weak and informative priors remain in clinically plausible ranges throughout. (These coefficient values are from a single representative draw with `set.seed(42)`; the robustness check below confirms they are typical rather than exceptional.)

Robustness Check: Repeated Subsampling

To confirm that these thresholds are not artefacts of a single random draw, we repeated the subsampling 100 times per n with independent seeds and recorded the proportion of runs in which the diffuse prior produced $|\hat{\beta}| > 10$ (Figure 3). At $n = 20$, **ht1** explodes in 31.1% of draws (and is undefined (no **ht** = 1 case sampled) in a further 26%). **smoke1** and **ui1** largely stabilize by $n = 40$ (explosion rates $\leq 3\%$), while **ht1** persists at 17.6% at $n = 40$ and reaches zero only at $n = 80$. The single-draw results above are therefore representative, not coincidental.

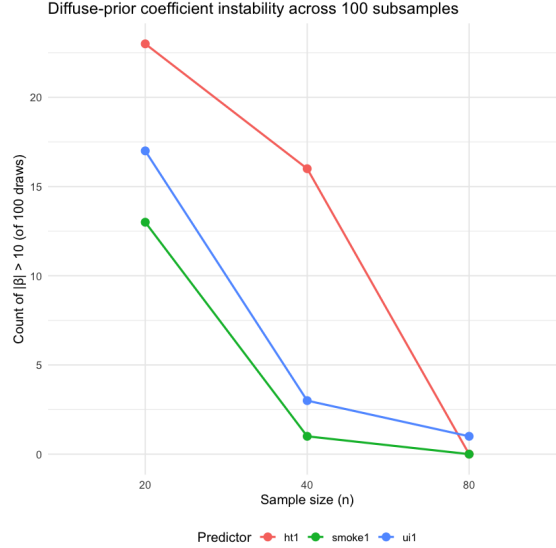


Figure 3: Subsamples (out of 100) where diffuse prior yields $|\hat{\beta}| > 10$. **ht1** is last to stabilize.

4.3 Predictive Performance

In-sample AUC is unreliable under separation; we use LOOIC and Pareto $\hat{k}$ diagnostics instead (Table 3, Figure 4). At $n = 20$ the diffuse prior’s AUC = 1.0 reflects overfitting, confirmed by 20 problematic $\hat{k}$ values and a spuriously low LOOIC of 8.3. The informative prior remains reliable throughout (1 problematic observation at $n = 20$) and achieves the lowest LOOIC at every sample size, including full data (Informative = 220.4, Weak = 223.3, Diffuse = 223.8).

n	Model	AUC	LOOIC	$\hat{k} > 0.7$
20	Diffuse	1.000	8.3	20
20	Weak	0.952	28.1	5
20	Informative	0.905	27.1	1
40	Diffuse	0.808	65.3	4
40	Weak	0.799	61.1	0
40	Informative	0.750	56.3	1
80	Diffuse	0.789	105.9	0
80	Weak	0.785	102.4	1
80	Informative	0.777	98.2	0

Table 3: In-sample AUC, LOOIC, and Pareto $\hat{k} > 0.7$ count by sample size and prior. At $n = 20$ the diffuse prior’s LOOIC and AUC are both unreliable.

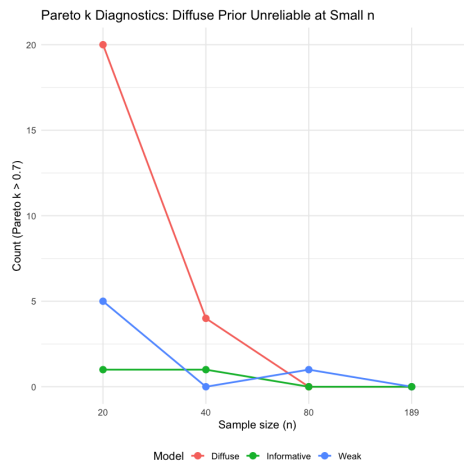


Figure 4: Pareto $\hat{k}$ diagnostics by sample size and prior. Values above 0.7 indicate unreliable LOO estimates.

5 Discussion

Two distinct mechanisms. Prior sensitivity operates through two channels: *separation stabilization*, where a diffuse prior lets coefficients diverge when no positive cases are sampled, and *gentle shrinkage*, where an informative prior pulls estimates toward clinical expectations even at full data. At $n = 189$, $\hat{\beta}_{ht1}$ shifts from 1.994 (diffuse, OR ≈ 7.3) to 1.311 (informative, OR ≈ 3.7), in the expected direction given published risk estimates [1, 2]; the posterior resists but does not eliminate prior influence.

Limitations. Results derive from a single 1986 dataset; generalizability requires validation. Subsampling without replacement may under-represent rare predictors in small draws. No prior predictive check was performed. At very small n , LOO is unreliable; k -fold cross-validation is preferable.

6 Conclusion

Prior choice matters most when data are scarcest. At $n = 20$ a diffuse prior yields unstable posteriors; by $n = 189$ all three priors are interchangeable. The stabilization threshold is governed not by sample size but by positive-case count: smoking (74) and uterine irritability (28) stabilize by $n = 40$, while hypertension (12) persists until $n = 80$, confirmed across 100 repeated draws.

For small clinical datasets: use a weakly informative prior by default, and treat rare predictors with particular caution.

References

- [1] Simpson, W.J. (1957). A preliminary report on cigarette smoking and the incidence of prematurity. *American Journal of Obstetrics and Gynecology*, 73(4), 807–815.
- [2] Shah, N.R., & Bracken, M.B. (2000). A systematic review and meta-analysis of prospective studies on the association between maternal cigarette smoking and preterm delivery. *American Journal of Obstetrics and Gynecology*, 182(2), 465–472.